

# Software Testing

Software testing is the process of checking if the quality, functionality, and performance of a software meets expectations and works as expected. Software Testing is a method to check whether the actual software product matches expected requirements and to ensure that software product is defect free. The purpose of software testing is to identify errors, gaps or missing requirements in contrast to actual requirements.

## Different Types of Testing

### 1. Black Box testing:

Black box testing means testing the software's functionality without any knowledge of its internal code structure. Testers focus on inputs and expected outputs to ensure the software meets requirements.

Example: Testing a calculator app by inputting various calculations and verifying correct results without examining the code.

### 2. White Box Testing:

White box testing is an approach where the tester examines the internal code structure and logic of the software. Testers have access to the source code and test specific paths, conditions, and loops.

Example: Checking the login functionality code of a banking app to ensure all branches (like handling incorrect passwords) work as expected.

### 3. Unit Testing:

Unit testing means testing individual components or functions of the software in isolation to verify that each part works as intended.

Example: Testing a single function like "Add to Cart" in an e-commerce app to ensure it performs correctly without involving other parts of the system.

### 4. Integration Testing:

Integration testing is the process of testing how different modules interact with each other to verify that they work together as expected.

Example: Testing the interaction between the shopping cart and payment gateway in an online store to ensure smooth data flow and successful transactions.

### 5. System Testing:

System testing means testing the complete, integrated system as a whole to verify that it meets specified requirements.

Example: Testing the entire flow of a flight booking app, from searching for flights to completing payment, to confirm that it works end-to-end.

### 6. Regression Testing:

Regression testing is the process of retesting existing functionalities after updates to confirm that recent changes haven't introduced new bugs.

Example: After adding a "Story Upload" feature in a social media app, retesting the "Photo Upload" feature to ensure it still works correctly.

### 7. Acceptance Testing:

Acceptance testing is the final testing phase where the client or end-users confirm that the software meets business needs and is ready for release.

Example: Librarians testing a library management system's core functions, like checking out and returning books, before it goes live.

**8. Stress Testing:**

Stress Testing means testing a system under extreme load conditions to determine the maximum load a system can handle before it begins to fail.

Example: Simulating heavy traffic on a news website during a major event to see if the system can handle the load without crashing.

**9. Usability Testing:**

Usability testing is the process of evaluating how user-friendly and intuitive the software is, focusing on the user experience.

Example: Observing how easily users can navigate a food delivery app to place an order, from browsing menus to completing payment.

**10. Validation:**

Validation means confirming that the software fulfills its intended purpose and meets customer needs, answering "Are we building the right product?"

Example: Validating that a hospital scheduling system allows doctors to book appointments that align with real-world clinic hours and patient needs.

**11. Verification:**

Verification is the process of ensuring that the software conforms to its specified requirements and design, answering "Are we building the product right?"

Example: Verifying that a payroll system accurately calculates wages and deductions based on predefined tax rules.

**12. Alpha Testing:**

Alpha testing is done by the internal team (developers or testers) at the developer's site before releasing the product to real users.

**13. Beta Testing:**

Beta testing is done by real users in a real-world environment to get feedback on performance, usability, and reliability.